Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:Simulate smoothing and sharpening operations on images using frequency domain filters.**	
Experiment No: 10	**Date:24/10/2026**	**Enrollment No:92400133041**

AIM: Simulate smoothing and sharpening operations on images using frequency domain Filters.

Theory: Smoothing and sharpening operations can also be performed in the frequency domain using filters such as the Gaussian filter and the Laplacian filter.

Smoothing in Frequency Domain:

To perform smoothing in the frequency domain, we apply a low-pass filter that attenuates high-frequency components. This helps to blur the image and reduce noise. One common approach is to use a Gaussian filter in the frequency domain.

Sharpening in Frequency Domain:

To perform sharpening in the frequency domain, we apply a high-pass filter that enhances high-frequency components. This amplifies the edges and fine details in the image. One common approach is to use a combination of a high-pass filter and the original image.

Flowchart:

The flowchart for the program will consist of the following steps:

Load the input image.

Convert the image to the frequency domain using Fourier Transform.

Apply the desired frequency domain filter (smoothing or sharpening) to the image in the frequency domain.

Convert the filtered image back to the spatial domain using Inverse Fourier Transform.

Display the original image and the filtered image side by side.

Save the filtered image (optional).

Program:-

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

img = cv2.imread('origi.png', 0)

# 2. Fourier Transform
F = np.fft.fftshift(np.fft.fft2(img))
```

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```
M, N = img.shape

# 3. Create frequency distance matrix
u, v = np.meshgrid(np.arange(N) - N // 2, np.arange(M) - M // 2)
D = np.sqrt(u**2 + v**2)

# 4. Low-Pass and High-Pass Filters
LPF = D < 30
HPF = D >= 30

# 5. Apply filters
smooth = np.abs(np.fft.ifft2(np.fft.ifftshift(F * LPF)))
high_pass = np.abs(np.fft.ifft2(np.fft.ifftshift(F * HPF)))
sharp = np.clip(img.astype(float) + high_pass, 0, 255)

# 6. Display images
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')

plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothening')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')

plt.show()
```

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Explanation:

We import the required modules, including cv2 for image processing, numpy for array operations, and matplotlib.pyplot for visualization.

The apply_gaussian_filter function performs smoothing by applying a Gaussian filter in the frequency domain. It uses the Fourier Transform (cv2.dft) and Inverse Fourier Transform (cv2.idft) functions to switch between the spatial and frequency domains.

The apply_sharpening_filter function performs sharpening by applying a high-pass filter in the frequency domain. It creates a mask that selectively allows high-frequency components to pass through.

In the main part of the program, we load the input image using cv2.imread and convert it to grayscale.

We apply the Gaussian filter to the input image by calling the apply_gaussian_filter function with a specified value of sigma. Adjusting sigma controls the amount of smoothing.

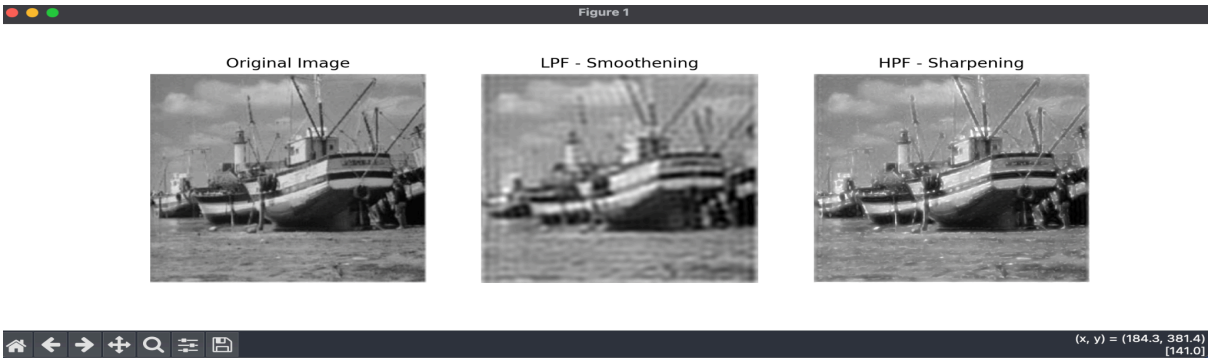
We apply the sharpening filter to the input image by calling the apply_sharpening_filter function with a specified value of strength. Adjusting strength controls the strength of sharpening.

The original image, smoothed image, and sharpened image are displayed side by side using matplotlib.pyplot.imshow.

The filtered images are saved to separate files using cv2.imwrite.

You can replace the image_path variable with the path to your input image. Also, you can modify the parameters sigma and strength based on your requirements.

OUTPUT:-



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Conclusion:

This experiment successfully demonstrated image processing in the frequency domain using the Fast Fourier Transform (FFT). By transforming the input image into its frequency components, continuous ideal low-pass and high-pass filters were applied based on spatial distance from the frequency origin. The Low-Pass Filter (LPF) successfully removed high-frequency details, yielding a smoothed, blurred image, while the High-Pass Filter (HPF) isolated edge transitions and high-frequency content. Combining the high-pass response with the original image effectively enhanced fine details and sharpened edge contrast, validating the practical effectiveness of frequency-based filtering techniques over spatial equivalents for global feature manipulation.

GITHUB LINK:-

https://github.com/SlockAhuja/DSIP.git